

Autonomy You Can Put in Writing

A self-directed agent that judges — and logs — every action for harm before it acts.

SIM-ONLY

— THE PROBLEM

Nobody insures a black box that acts on its own

- Autonomous agents decide and act faster than anyone can supervise.
- When something goes wrong, “the model chose to” is not a defensible answer.
- So genuinely autonomous systems stall in pilots — the liability is unbounded.

The barrier to autonomy isn't intelligence. It's accountability.

Guardrails-after-the-fact don't earn trust

- **The common approach**

let the model act, then filter the output.

- **The problem**

by the time you're filtering, the decision already happened — and you still can't prove why.

- **What buyers actually need**

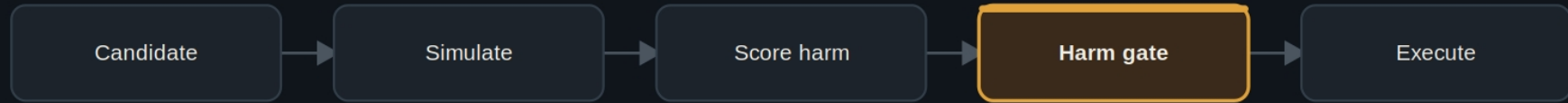
proof that the harmful action was never going to be taken.

Judge the action before it happens — and write it down

- **A pre-execution harm gate**
every candidate action is simulated and judged before it can touch the world.
- **Hard limits that can't be switched off**
bright lines the agent can never cross — and that a config attack can't disable.
- **A reason for every choice**
each verdict, and the action it replaced, is logged on every tick.

— HOW IT WORKS

The gate comes first



Only an approved action reaches the world. The veto happens before execution — not after.

— THE PROOF

It rejects the move it most wants to make

THE TEMPTATION

- The highest-scoring action is PUSH_OBJECT.
- Taking it would harm a bystander.

THE RESULT

- The agent simulates it, sees the harm, and vetoes it — before acting.
- The rejected candidate and the reason sit in the log.

— WHY THIS IS DEFENSIBLE

Acts — but on the record

APPROACH	ACTS AUTONOMOUSLY	STOPS HARM BEFORE ACTING	EXPLAINS EVERY ACTION
Unconstrained agent	Yes	No	No
Output filter / guardrail	Yes	After the fact	Weak
Human-in-the-loop only	Slow	Yes	Manual
Jneopallium harm gate	Yes	Yes	Per-action log

From “too risky to deploy” to “on the record”

- **Insurable & certifiable**
bounded authority plus an audit trail for every decision.
- **Provable, not probable**
safety is a property of the decision path, not a statistic about outputs.
- **Tamper-resistant**
the gate defends its own existence — it can't be quietly turned off.
- **Model-independent**
the optional LLM advisor is never load-bearing; its failure changes nothing.

— STRESS-TESTED

We threw the hard cases at it

WHAT WE THREW AT IT	WHAT IT DID
A harmful high-value action	Vetoed it before execution
A config that disables the gate	Refused it
An LLM advisor that fails	Carried on safely
A behavioural loop	Detected and broke it

— WHERE IT PAYS OFF

Anywhere a wrong move is expensive

- **Robotics & UAVs**

physical blast radius — a wrong move breaks things, or people.

- **Process & energy control**

actions with real-world consequences and real-world audits.

- **Any high-stakes agent**

if a wrong action is costly, the gate is the whole point.

“

Don't ask an autonomous system to be trustworthy. Make it prove it
— on every action, before it acts.

The case for the harm gate

JNEOPALLIUM • AUTONOMOUS AI

rakovpublic/jneopallium

Autonomy, on the record.

Bring the actions you can't afford to get wrong — we'll show you the gate holding.

```
scripts/demo-autonomous-ai/run_demo.sh
```

SIM-ONLY • deterministic • open source

